VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Reg. Number:

Continuous Assessment Test 2 (CAT) – Oct 2024 -25

Programme	:	B.Tech - BCE, BAI, CPS, BRS	Semester	:	Fall 2024-25
Course Code & Course Title	:	BCSE301L & Software Engineering	Class Number	:	CH2024250102293, CH2024250100968, CH2024250100976, CH2024250101320, CH2024250102297, CH2024250101319, CH2024250100972, CH2024250101387, CH2024250101385, CH2024250101321, CH2024250102296, CH2024250102451
Faculty	:	Dr.Jayanthi.R, Dr.S.Brindha, Dr.Suganeshwari G, Dr.N Ganesh, Dr.G.Saranya, Dr. Smrithy G S, Dr.N M Elango. Dr. Rama Prabha K. P, Dr.Praveen Joe I R, Softya Sebastian, Berin Shalu S, Dr. SK. Karthika	Slot	:	G2+TG2
Duration	:	90 Minutes	Max. Mark		50

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information.

Answer all questions

Q. No	Sub Sec.	Description	Marks
1.		An E-Commerce Platform enables the buying and selling of products across various regions. The system is designed to handle numerous tasks, such as managing product listings, processing orders, overseeing logistics, providing customer service, and facilitating payment transactions. This platform will cater to diverse users, including customers, sellers, logistics partners, and administrators.	10
	a)	Design the use case anatomy for the E-Commerce Platform with its main Use case diagram. (5 Marks)	
	b)	Design a Class diagram for the above scenario. (5 Marks)	
2.		ABC is an E-Learning Platform designed to provide an engaging, interactive, and collaborative learning environment for both students and educators. However, the current software version falls short of offering flexibility, personalization, and effective tools for student-instructor interaction. To overcome these limitations, a new version of the E-Learning Platform will be developed using the Agile model. This approach will allow the project team to deliver iterative updates based on user feedback and the	10

	a)	evolving needs of the educational community. In the context of developing the new version of ABC's E-Learning Platform using the Agile model, how would you manage the gathering of requirements from diverse users such as students, instructors, and administrators? (5 Marks)	
	b)	Discuss the Process of Refining two different user stories for the E-Learning Platform in Agile Requirements Management. (5 Marks)	
3.		Design a Smart Traffic Management System (STMS) for a futuristic city to reduce congestion and optimize road usage in real-time. The system collects data from traffic sensors, GPS systems, and camera feeds, pre-processes it to filter noise and identify patterns, and uses it to optimize traffic flow by adjusting traffic lights and rerouting vehicles. It also detects accidents or heavy congestion and notifies emergency services. Create a transformation mapping to support the transition from logical Data Flow Diagrams (DFDs) to a system architecture for the STMS.	10

4.		Consider the following **C** code snippet:
		`int main() {`
		`  int num1, num2;`
		`  printf("Enter two integers: ");`
		`  scanf("%d %d", &num1, &num2);`
		
		`  if(num1 == num2) {`
		`  printf("Result: %d = %d\n", num1, num2);`
		`  } else if(num1 > num2) {`
		`  printf("Result: %d > %d\n", num1, num2);`
		`  } else {`
		`  printf("Result: %d < %d\n", num1, num2);`
		`  }`
		`  return 0;`
		`}`
			8

	a)	Draw the control flow graph for the above code (4 marks)	
	b)	Calculate the Cyclomatic complexity from the graph (4 marks)	
5.	a)	List the different types of regression testing that are relevant to the Smart Traffic Management System (STMS) described in **Question No. 3.** (3 Marks).	12
	b)	Discuss how different types of regression testing apply to the Smart Traffic Management System features like traffic signal control and incident detection, **Ref. Question. No. 3** (9 Marks)	

**********All the best **********